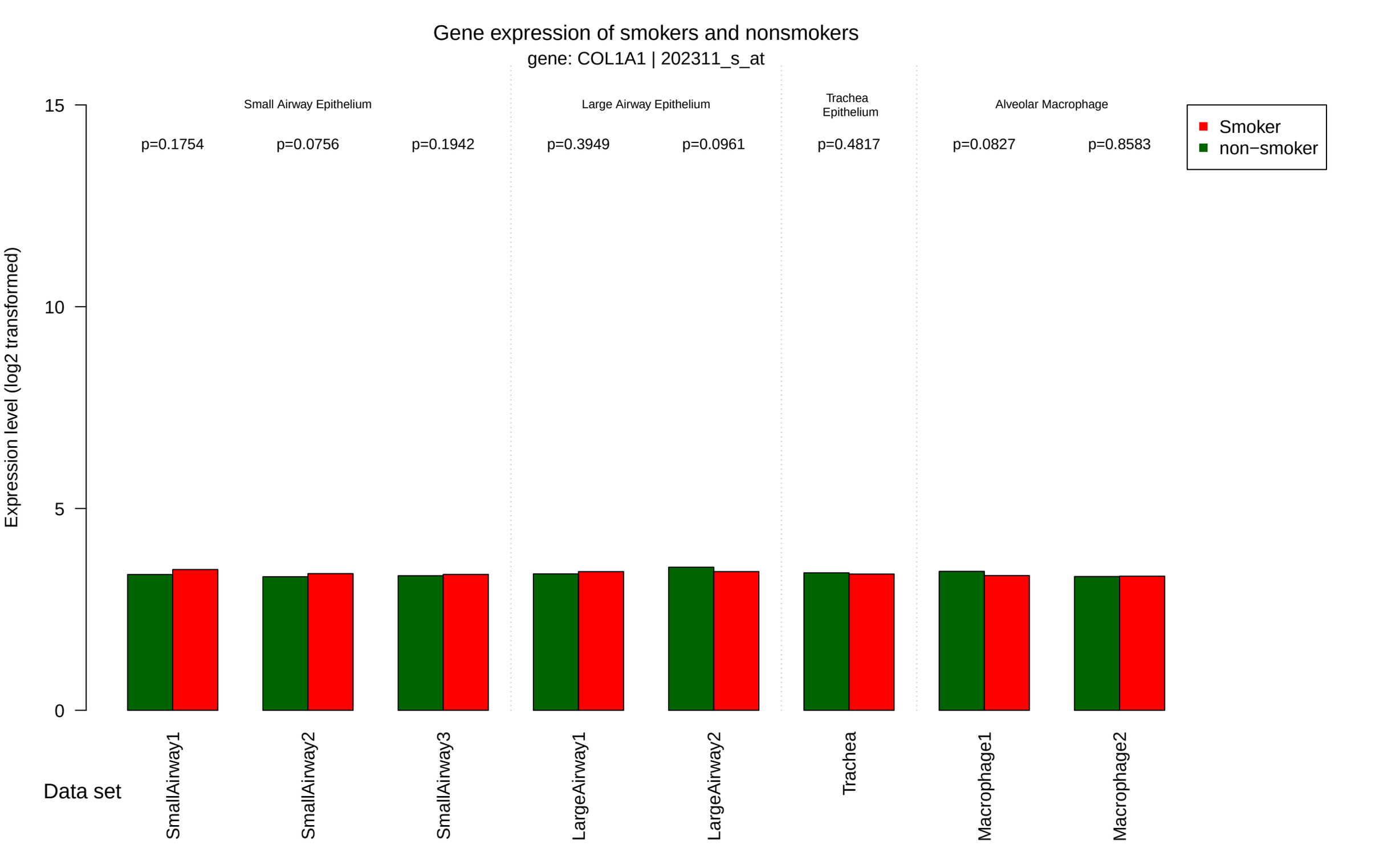
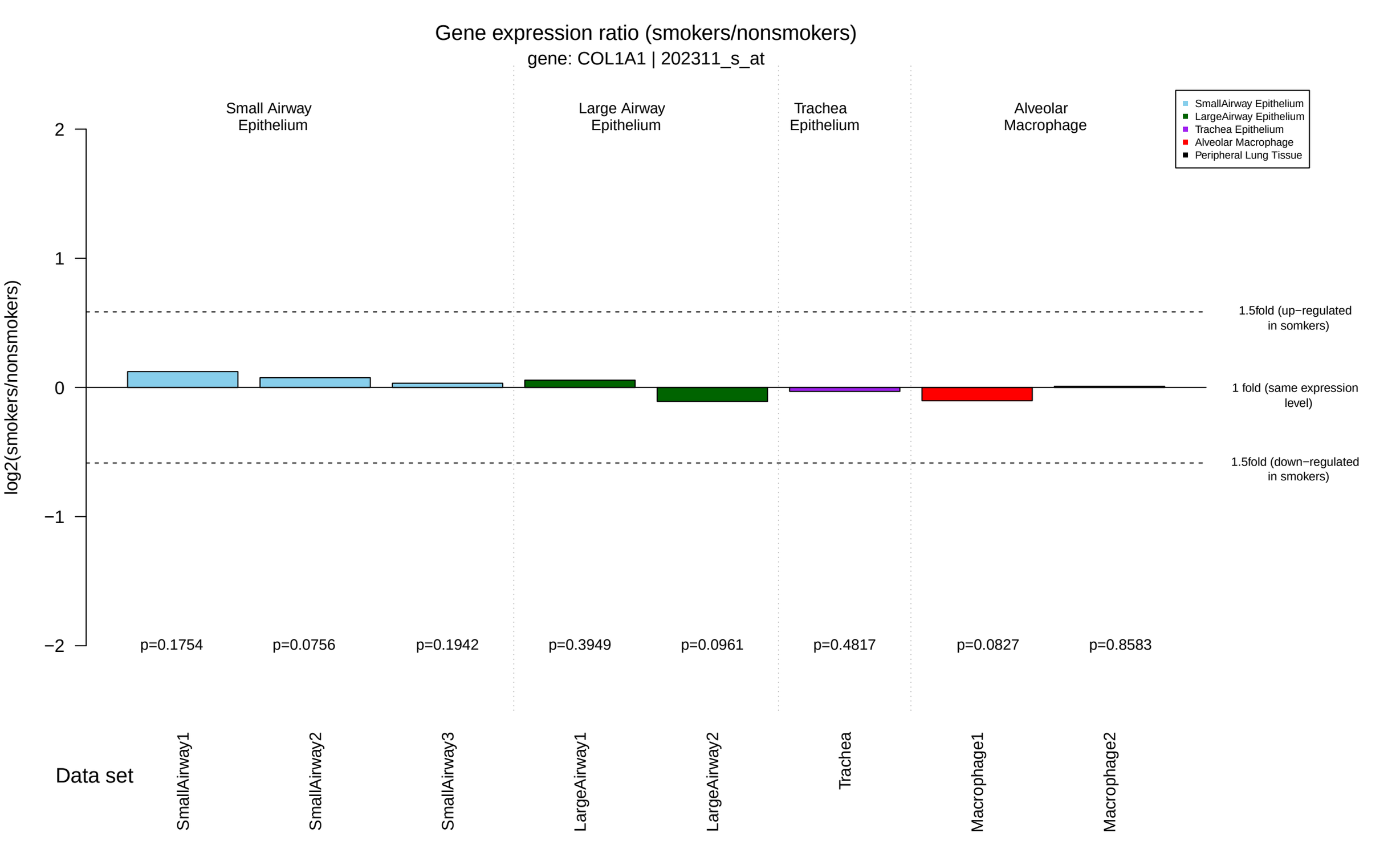
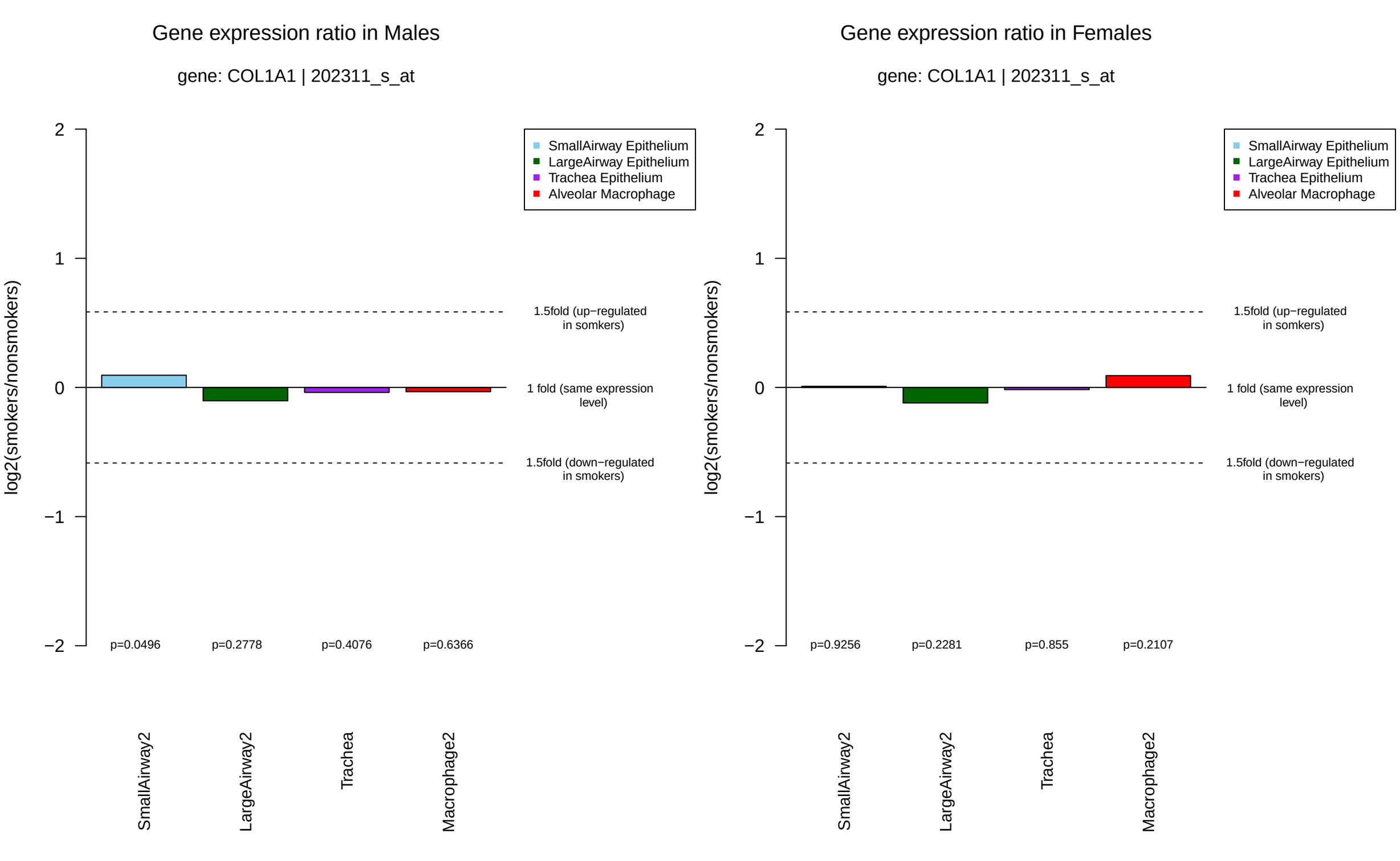
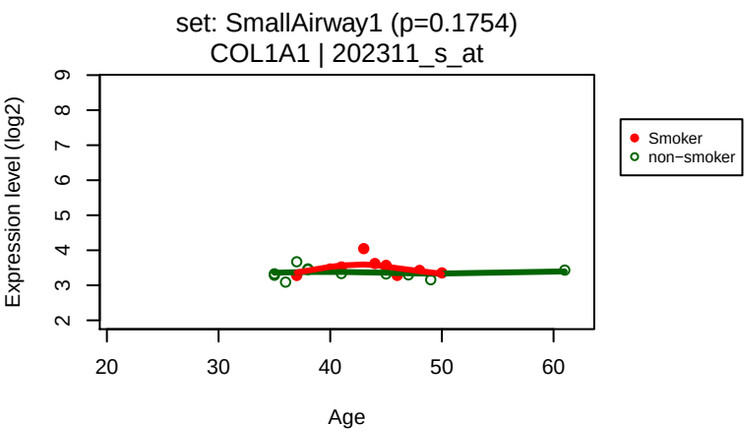
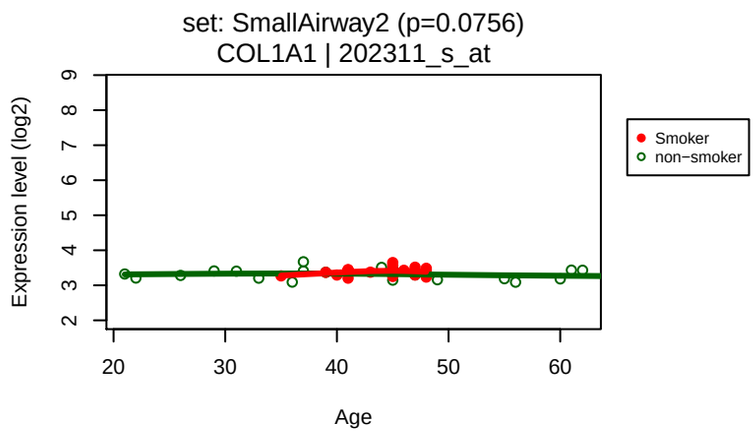
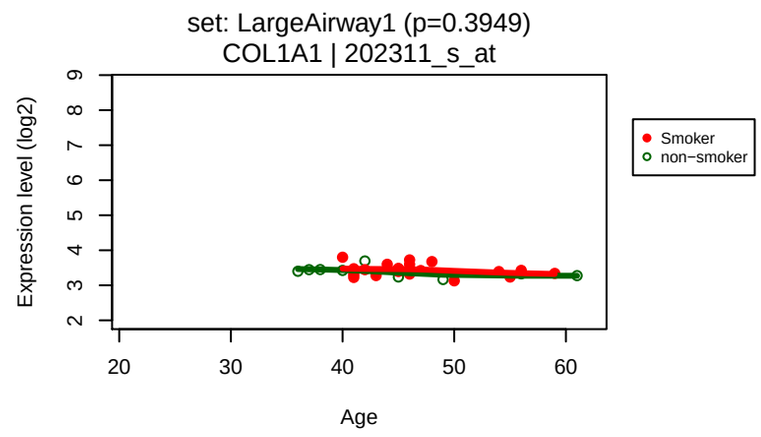

Gene expression of smokers and nonsmokers
gene: COL1A1 | 202311_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1754
p=0.0756
p=0.1942
p=0.3949
p=0.0961
p=0.4817
p=0.0827
p=0.8583
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

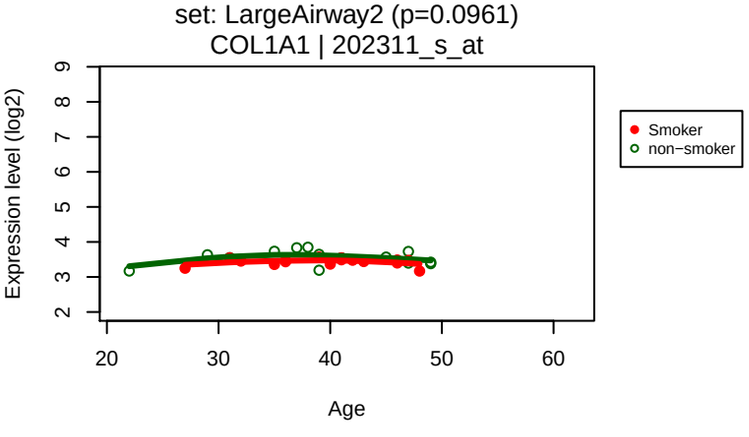

Gene expression ratio (smokers/nonsmokers)
gene: COL1A1 | 202311_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1754
p=0.0756
p=0.1942
p=0.3949
p=0.0961
p=0.4817
p=0.0827
p=0.8583
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

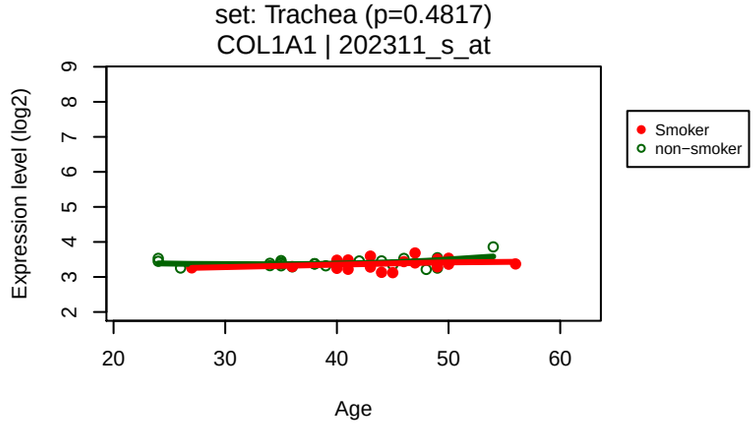

Gene expression ratio in Males
gene: COL1A1 | 202311_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0496
p=0.2778
p=0.4076
p=0.6366
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL1A1 | 202311_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9256
p=0.2281
p=0.855
p=0.2107
SmallAirway2
LargeAirway2
Trachea
Macrophage2

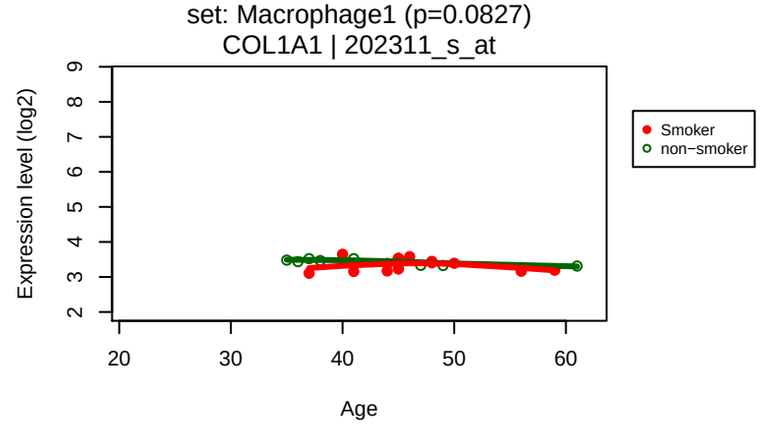

set: SmallAirway1 (p=0.1754)
COL1A1 | 202311_s_at
Expression level (log2)
Age
Smoker
non−smoker

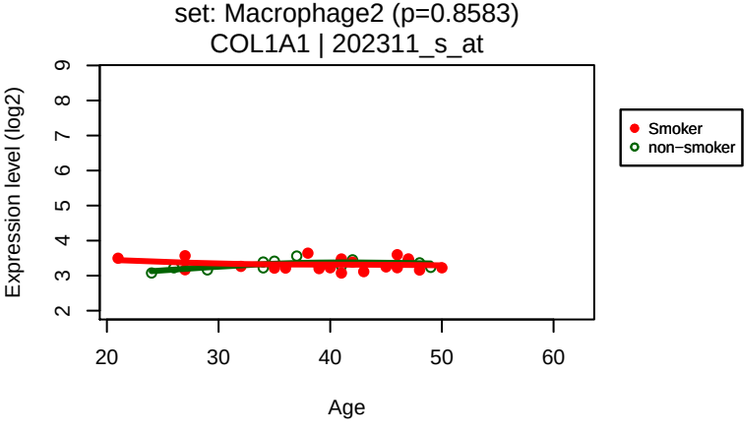

set: SmallAirway2 (p=0.0756)
COL1A1 | 202311_s_at
Expression level (log2)
Age
Smoker
non−smoker

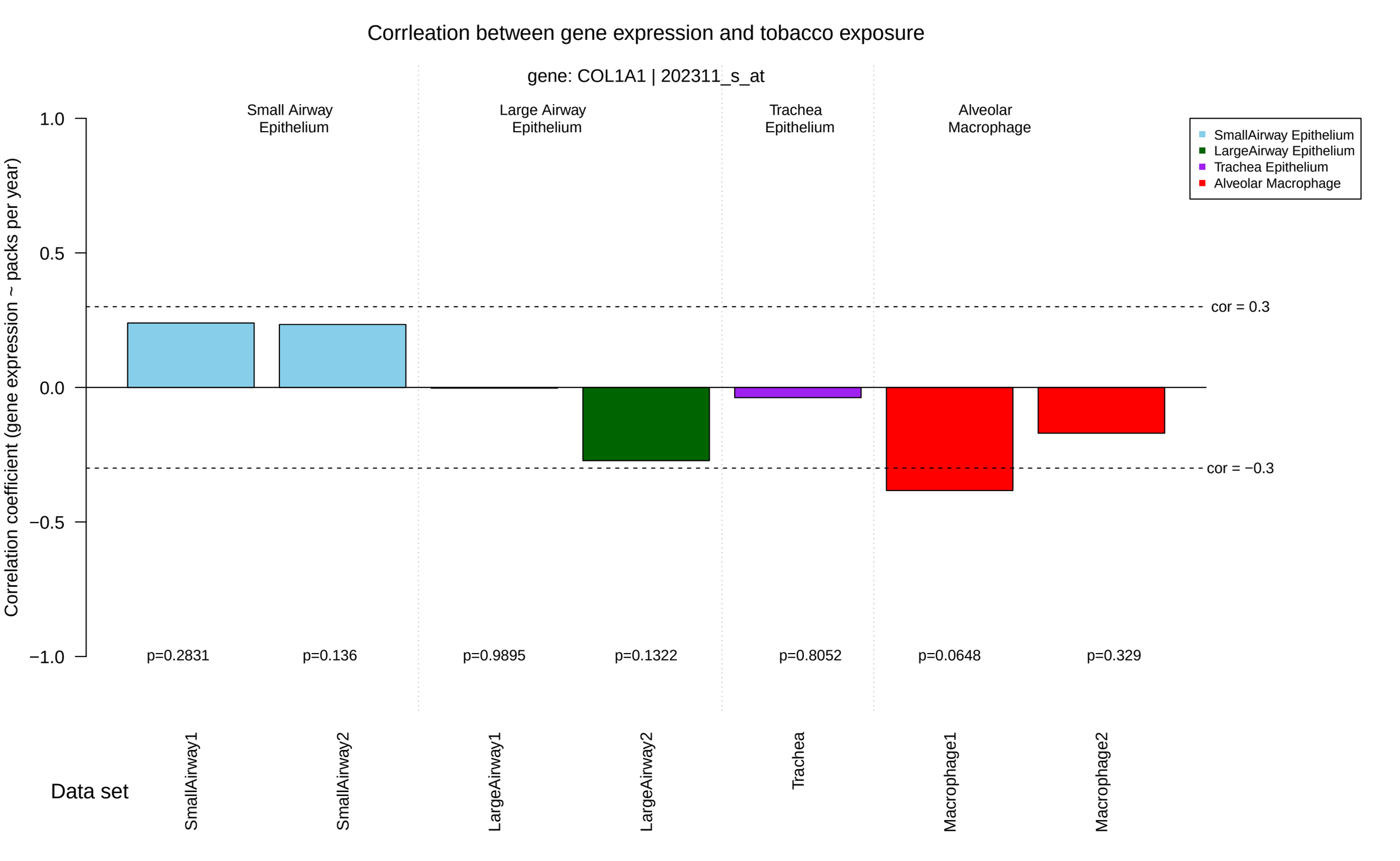

set: LargeAirway1 (p=0.3949)
COL1A1 | 202311_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0961)
COL1A1 | 202311_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4817)
COL1A1 | 202311_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0827)
COL1A1 | 202311_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8583)
COL1A1 | 202311_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL1A1 | 202311_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2831
p=0.136
p=0.9895
p=0.1322
p=0.8052
p=0.0648
p=0.329
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2